Review of Document Management System in Secured and Cloud Environments

Siya Mehta
Bsc IT
12308033
Lovely Professional University
Phagwara, Punjab

Abstract—**The review paper aims to analyze developments in the domain of digital governance, digital identity systems, and service infrastructures based on smart lockers using findings of various national and international studies. In particular, the paper discusses that large-scale implementation of digital identity initiatives is considered one of the major goals due to their potential to improve service delivery, efficiency, and transparency; at the same time, these projects are associated with certain challenges relating to interoperability, data security, exclusion risk, and potential of surveillance. Another aspect discussed within this research is an impact of Digital Public Infrastructure (DPI) on provision of seamless access to service in different areas including governance, health care, e-commerce, etc., as well as on inclusivity and resilience. Finally, the paper considers recent developments in the field of automated parcel lockers and smart lockers as an integral part of the last mile delivery process and its main advantages and disadvantages related to logistics and infrastructure development. In this regard, through comparative analysis and discussion of findings, the paper reveals certain aspects as crucial for efficient and effective implementation of innovations, including federated architecture, interoperability standards, citizen-consented approach, and integration of service portals. Also, certain problems and opportunities can be revealed, particularly related to policy and law framework in relation to data protection.**

Keywords—Digital Identity, E-Governance, DPI, Interoperability, Data Protection, Smart Lockers, Service Delivery, Digital Transformation

I. Introduction

With the emergence of digital governance and digital identity management systems, the nature of public administration and governance has started changing[1][3], as governments find the opportunity to improve their effectiveness, transparency, and citizen involvement in governance. This change is enabled by the idea of digital identity as a legibility infrastructure that allows governments to identify, verify, and manage the identities of citizens using various services offered by them. Even though digital governance and identity management offer significant advantages in terms of the provision of government services and efficiency in governance, they also pose serious risks, especially regarding surveillance, exclusion, and protection of citizens' personal data[5].

Some examples of digital identities are seen in nations like Nepal and India, which reveal opportunities as well as restrictions. For instance, the National Identity Card (NID) system in Nepal is characterized by the high percentage of card registration[7][8]. It signifies that this type of technology enjoys considerable support from Nepalese citizens when used initially. Nevertheless, there exist severe obstacles in integrating and interlinking these cards with other government operations because of the lack of uniform architecture and law. It means that apart from being used widely, digital identity technologies have to be supported by robust backend infrastructure and legislation. On the contrary, India has made significant advances through various projects under the Digital India scheme, including Aadhaar, Digi Locker, and e-EPIC.

An important enabling factor that facilitates such a shift is called the Digital Public Infrastructure (DPI). DPI acts as the infrastructure that connects identity, payments, and service delivery interfaces. Using DPI, governments have developed scalable and interconnected governance systems which will be able to address a large populace. This can be seen from the implementation of the Apuna Sarkar portal which offers access to many governmental services through a single platform, greatly benefiting the underprivileged areas[8]. In spite of these developments, there are obstacles such as the issue of digital divide, lack of digital literacy, obsolete legal framework, and issues related to data security.

Conversely, the fast expansion of global e-commerce has raised new issues of logistical nature, especially in relation to last-mile deliveries. The conventional means of delivering packages tend not to satisfy the growing needs in terms of efficiency, causing a series of problems like delays and high costs. To solve this issue, a number of innovations has been developed, including automated and smart lockers. They provide a convenient, flexible, and safe way for receiving orders by ensuring that packages will be collected according to the consumer's schedule without any failed delivery attempts.

All in all, the coming together of digital identity systems, DIP, and logistics innovations can be viewed as an important milestone towards developing more efficient, transparent, and citizen-oriented systems. Yet, a truly sustainable and inclusive approach to digital transformation necessitates further progress in technology and policy frameworks alike, making sure that everyone in society has access to digital innovations.

II. LITERATURE REVIEW

There is an increasing realization about the significance of digital identity systems, smart infrastructure, and the use of technology for the provision of services in order to increase efficiency and accessibility. One of the emerging themes is that of federated digital identity systems that help in improving the level of security, building the trust of users due to the ability of auditing, and integrating more than one public service at a go. Additionally, convergence of digital identity helps with authentication, avoiding duplications, and efficiency through paperless systems.

Regarding the field of logistics and last-mile deliveries, one example that shows remarkable progress in terms of accessibility, cost-effectiveness, and sustainability is the introduction of smart lockers especially when combined with mobile apps, Internet of Things, and public transportation. This technology not only increases customers' convenience through 24/7 access and efficient parcel management but also helps reduce congestion and expenses. Moreover, financial and strategic analysis suggests that investment into smart infrastructure, for instance, intelligent locker networks, will help generate growth.

In addition, the use of digital public infrastructure and e-governance platforms is vital for the transformation of the process of delivering public services in that they can foster transparency, accountability, and a citizen-oriented approach. The technologies used may include cloud computing, artificial intelligence, blockchain technology, and APIs, making the process of integration between different services possible and promoting inclusiveness as well as rural penetration of new services. Yet, despite these technological advances, several issues still exist. Some of the main barriers to digital transformations are related to the problems associated with data protection and security, risks of digital exclusion among disadvantaged communities, lack of digital literacy, and infrastructural restrictions. The majority of studies are mainly based on secondary data or theoretical modeling and do not offer any geographical diversity of cases under study. In addition, cost issues, complexity of integration, difficulties with scalability, and constantly changing regulatory environment may be seen as other significant obstacles.

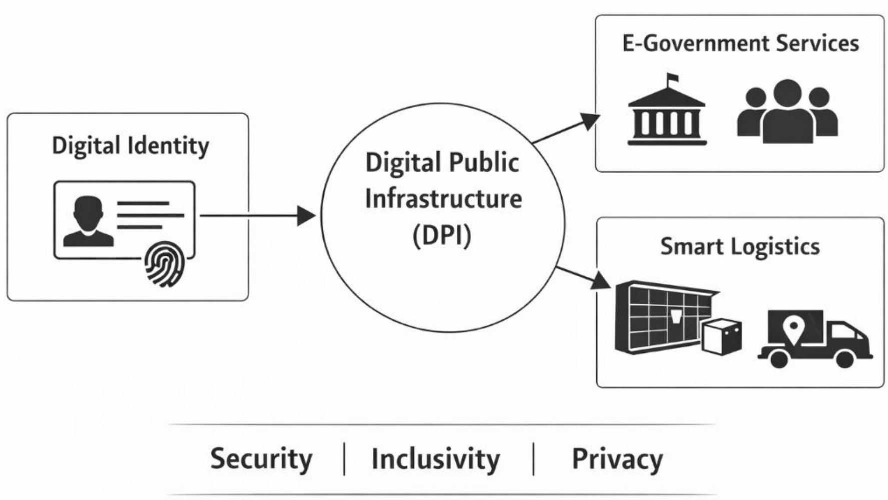

Figure 1: Digital Identity and Digital Public Infrastructure (DPI) based E-Governance and Smart Logistics Framework

III. METHODOLOGY

It is clear that the methodologies used in these researches have been meticulously chosen to ensure comprehensive analysis in all its dimensions. Firstly, in the first research paper, a mixed-methodology is employed, which consists of an extensive use of secondary data and the study of various countries around the world to analyze the effectiveness of citizen engagement in the process of digitalization. Secondly, the second research paper makes use of a highly sophisticated GIS-based spatial analysis methodology in assessing the efficiency of smart parcel lockers' accessibility and equity. Lastly, in the third research paper, a thorough qualitative methodology of case studies is used to assess the transformational impact of Digital Public Infrastructure (DPI) on governance via the Apuni Sarkar Portal.

IV. APPLICATIONS

These papers have adopted an approach involving several sophisticated technological and analytical tools, platforms and resources in order to conduct an in-depth analysis. The technologies used include GIS (Geographical Information Systems) and Google Earth technologies for purposes of spatial mapping, accessibility, and infrastructural analysis. Important data and statistical resources include U.S. Census Bureau's American Community Survey, County Business Patterns, UN E-Government Survey, OECD studies, World Bank databases, among others. Government digital resources, including the Digital India portal, Apuni Sarkar portal, and DigiLocker, are crucial for delivering the services. Different analytical methods, including Gini coefficient for equity analysis, thematic coding, among others, and the use of survey and feedback mechanisms for analyzing citizen participation are some of the analytical methods that are utilized.

In this case, the research paper mainly uses automated parcel locker systems as a smart logistics technology in cities, specifically in the last mile delivery of e-commerce operations. APLs are used in self-service parcel collection and delivery services where the customers are able to retrieve and deposit their parcels anytime of the day without any face-to-face interaction with humans. The paper mentions how applications can be used together with the parcel lockers for tasks such as parcel tracking, unlocking, notifications, and contactless delivery, among others, making the process easy and efficient. Apart from that, the apps can be extended to be used for smart logistics within cities by enhancing the efficiency of the delivery process, minimizing congestion, and ensuring the sustainability of the urban transport system. Lastly, in the paper, the Kano Model and surveys have been used to analyze various aspects of customer satisfaction and service quality attributes.

The research paper uses the digital governance and e-governance technologies as technological applications to improve public administration and citizen engagement in India. This is based on the application of Information and Communication Technologies (ICTs) in delivering public services through digital service portals, mobile apps, and citizen participation websites. The ICT applications that will be applied in this research are citizen service delivery applications (citizen services such as online certificate issuance, pension services, and tax services), digital identity and documents issuance platform (for instance, DigiLocker), integrated mobile governance applications (UMANG), and digital payments platforms such as UPI. In addition, the research uses platforms such as citizen engagement sites (MyGov). This will ensure citizen consultations and engagement online. These applications will help deliver effective services; foster transparency; and promote real-time communication between the government and citizens. The research paper uses different important digital tools and platforms to be a part of the Indian digital identity framework. Such tools include Aadhaar (UIDAI) used for identification and authentication based on biometrics, e-EPIC or the Digital Voter ID for secure voter authentication, DigiLocker for the storage and retrieval of important digital records, and the MY Bharat Portal for citizen-centric services. These tools are linked to each other using various methods of authentication such as OTP (One Time Password) and biometric authentication. This helps facilitate easy data interoperability and exchange between systems, real-time verification of identities, paperless governance, and more efficient provision of administrative and electoral services.

V. CHALLENGES

1. The implementation of the NID system in Nepal has to deal with an obvious "integration gap" due to the low level of card collection (just 13% by mid-2025) and inadequate service connectivity among fewer than ten national services. Such a delay is prompted not only by the architectural but also by the legal and organizational problems in the process of integration. For instance, some organizations such as Land Revenue Offices oppose NID since having unified records may reveal discrepancies in the titles of individuals, while there is no incentive for Inland Revenue Department to integrate the identifiers because it charges the citizens for registering their PAN cards separately[1].

2. The growing demand for retail e-commerce services which operates through online platforms creates major logistical problems for urban areas because it impacts the delivery process which must find quick and cost-effective ways to deliver products to final customers. The "last-mile" process of delivery presents major challenges which require organizations to implement effective solutions that use automated parcel lockers (APLs) for handling delivery failures and tracking lost or incorrectly delivered packages. Service providers face challenges in their market differentiation efforts because they struggle to find and rank the quality factors which directly impact customer satisfaction[2].

3. The paper identifies major obstacles to digital governance implementation because 37% of Indian villages currently lack access to reliable high-speed broadband internet. The country suffers from a critical digital literacy gap which prevents 75% of Indians aged 15 and above from using computers or the internet. Linguistic barriers create a major challenge because 88% of users prefer regional languages while most portals remain in English or Hindi. The rapid digitalization process has made organizations more open to cyber incidents which resulted in 13 lakh reported cases during the 2022 period[3].

4. The paper identifies several important structural obstacles which continue to exist because of digital efficiency improvements yet they require special attention to two main issues which involve data security and user privacy protection within the unified system. The country faces two major challenges which include the need to protect vulnerable groups from digital exclusion and the existing problem of low digital skills among its population. The federal system faces difficulties because it needs to achieve effective coordination between central and state governments to maintain the digital identity ecosystem for extended periods[4].

5. The fixed parcel lockers which operate as permanent installations create problems for customers who live far from the lockers because they cannot reach the lockers and because their needs fluctuate throughout the day. Although mobile parcel lockers (MPLs) provide customers with increased movement options, the system requires extensive financial investment to establish and run the system while facing challenges to find appropriate areas for effective pickup operations. The existing systems encounter major problems because they experience low usage of their lockers and face challenges from legal regulations and they need to spend money to get occupancy permits. Public transport networks require better technology integration because transit agencies and logistics providers need to work together to solve existing coordination problems[5].

6. The primary challenge which has been identified occurs because traditional mail volumes have decreased significantly because people now use digital communication methods which reduced letter volumes from 2019's 1,140.4 million items to 2023's 680.2 million items. The Mail unit experienced severe profitability pressures because its EBIT decreased from €42.9 million to zero between the two time periods. The company needs to maintain a nationwide retail network and daily mail carrier routes because of its universal service concession which creates high fixed costs and requires many workers. The company must deal with operational challenges because it needs to deliver excellent service while it expands its business operations throughout Spain[6].

7. The research identifies multiple essential obstacles which prevent Uttarakhand from achieving complete digital governance implementation because of the existing digital divide which separates people with different internet access and smartphone ownership rates between urban and rural or mountainous regions. The adoption of services faces additional challenges because of two structural problems which include administrative inertia and different departmental dedication levels and the public's limited understanding of digital systems. The portal establishes a "faceless" operation system yet numerous citizens continue to depend on Common Service Centers (CSCs) as intermediaries which create access problems because these centers exist in remote areas with unreliable services[7].

8. The paper identifies several significant obstacles to Digital Public Infrastructure (DPI) which successfully requires implementation through two existing legal frameworks and the widespread resistance to change that exists in developing countries. A major concern is the prevalence of digital exclusion which exists because 850 million people worldwide lacked formal identification in 2021 and 5 billion people lived in countries that lacked secure electronic IDs. The risks which involve data security and privacy protection together with the potential for excluding marginalized groups present critical challenges which policymakers need to address[8].

VI. LIMITATION

1. One of the weaknesses pointed out in the research paper concerns the fact that the study has recognized its own limitations including lack of primary data collected from the field and the "small-N" comparative case study approach, which involved comparing six different national digital identity systems. As cases were picked because of their accessibility and not randomly chosen, it was acknowledged that causal interpretation is not possible since design-based hypotheses, although interesting, could have been driven by reporting bias or recentness bias. Moreover, sectors such as health care and education were excluded from the study[1].

2. The study notes that not all parcels can fit within the specified dimensions of standard APLs, requiring alternative delivery methods for oversized items. The research concentrated on 468 respondents from the Silesian Metropolis in Poland who mostly belonged to the 18-to-34 age group which restricts the ability to apply the results beyond this age range and other geographical areas. The study demonstrates that present users consider certain features as "indifferent" which includes environmental awareness and temperature control yet these features will become more important for user satisfaction in upcoming times[2].

3. The study methodology needs to use secondary data which does not provide access to contemporary information and lacks the ability to document the particular conditions which affect people who experience digital exclusion thus creating a tendency to report positive results. The research reveals that the fast technological advancements will make its results obsolete within a short period. The research findings face obstacles because India and the studied global leaders have major differences in their population size and development status and administrative systems[3].

4. The paper establishes its findings through secondary data analysis which includes policy documents and institutional reports covering the period from 2015 to 2025. The study demonstrates that digital efficiency depends on how well local systems execute their operations. The study shows that anti-corruption improvements which result from the research will face authentication challenges which remain unmitigated. The study demonstrates that digital identity systems require strong regulatory controls and institutional monitoring systems to deliver their complete advantages[4].

5. The study establishes its results through the premise that sufficient locker-equipped buses must operate on the designated routes. The research results show specific applicability to Portland, Oregon, which demonstrates that other regions with less developed transit systems need dedicated funding to achieve identical coverage and equity improvements. We Pick service area expansion enables more people to access services through walking, but the service population growth remains limited because the system extends into less populated suburban and rural areas. The paper establishes that we Pick functions as a specific solution which works together with fixed lockers to provide better service in particular situations[5].

6. The document investigates the growth possibilities of the Express & Parcels (E&P) segment while pointing out that the company depends excessively on its traditional postal services which now experience decreasing demand. The logistics segment has experienced a major decline because its current contribution to recurring EBIT decreased from 78.5% in 2019 to 49.3% in 2024. CTT has experienced a decline in its postal sector market share which started at 83.3% in 2018 and reached 76.6% by 2023. The research indicates that economic fluctuations along with strict EU mail regulations create financial restraints for the company which results in increased operational expenditures[6].

7. The research demonstrates its boundaries through its use of qualitative single-case study which examines Uttarakhand and lacks ability to extend its results to other states with distinct geographic and socio-economic characteristics. The paper utilizes primary data from interviews and direct observation, yet the actual effectiveness of grievance redressal systems depends on local administrative capacity, which creates variability in the study's efficiency assessment because departmental differences exist. The platform's sustainability requires both ongoing institutional funding and sustained adherence to established policies, which represent external requirements that exist outside the portal's direct technological framework[7].

8. The study uses existing case studies and literature instead of conducting new empirical research to evaluate its conceptual and policy-related part. The study recognizes that digital readiness varies across European Union regions which makes it impossible to use a standard approach. The paper shows that many countries Today have insufficient

physical infrastructure and their citizens lack basic digital skills which both limit successful implementation of their designed DPI systems[8].

VII.FUTURE SCOPE

The future scope of this study requires researchers to conduct both practical studies and extended research studies which will test existing theoretical models while they work to enhance digital system deployment. The effectiveness of federated architectures and digital identity systems and service delivery platforms needs assessment through longitudinal and comparative studies that will evaluate their performance in various regions. Advanced technologies like Artificial Intelligence (AI) and blockchain and big data analytics will help governance and logistics systems become more transparent and efficient and personalized and decision-making will improve. Future developments should create integrated platforms which connect various services and departments and maintain strong data protection and privacy and regulatory compliance. Digital inclusion requires better accessibility and digital literacy programs and infrastructure development to prevent marginalized populations from facing exclusion. Last-mile delivery efficiency and customer satisfaction in logistics can be improved through smart and mobile parcel lockers and AI-based demand forecasting and optimized delivery networks. Policy-level improvements which include simplified regulations and public-private partnerships and international cooperation are necessary to support scalable and sustainable digital ecosystems. Future research should create systems which adapt to changing needs while protecting security and helping citizens achieve their goals through systems which integrate technological development with social and legal and economic needs.

VIII.CONCLUSION

The review indicates that digital identity systems together with Digital Public Infrastructure (DPI) and smart service technologies create a complete system which improves governance and service delivery while making services more accessible to people. The research shows that security needs in identity systems require organizations to adopt federated systems which use interoperable architectures that work with open standards. The implementation of digital identity frameworks and unified service portals enables better administrative operations which create higher levels of transparency and democratic engagement according to evidence from Nepal and India. Digital governance success depends on the implementation of new technologies together with the establishment of institutional abilities and the creation of inclusive policies and the development of user capabilities. The logistics industry uses its advanced delivery systems which include automated parcel lockers and smart parcel lockers to improve last-mile delivery operations while increasing accessibility and customer satisfaction among those who face disadvantages. The expansion of e-commerce and digital services emphasizes the need to integrate digital systems with physical systems in order to deliver complete user experiences. Organizations need to establish a sustainable method which tracks their efficiency while securing data and respecting regulatory requirements and providing equal access to their resources. The research demonstrates that digital ecosystems of the future need to build security systems which include all users while creating services that protect user privacy rights and safeguard customers against harm.

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